1	2	3	4	5	6	Sum
/20	/20	/20	/20	/20	/20	/100

Name ______________________ Nov. 7, 2003 Sunday

Department _________________

SID _______________________

University Physics

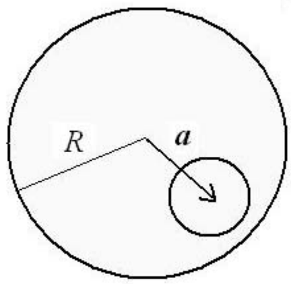
Midterm Examination

School of Intensive Instruction in Science and Arts, Nanjing University

Physical Constants

Electron charge	$e = 1.60\times10^{-19}$ C
Dielectric constant of free space	$\varepsilon_0 = 8.85\times10^{-12}$ F/m
Permeability of free space	$\mu_0 = 4\pi\times10^{-7}$ N/A^2
Vacuum speed of light	$c = 3.00\times10^{8}$ m/s
Electron mass	$m_e = 9.11\times10^{-31}$ kg
Proton mass	$m_p = 1.67\times10^{-27}$ kg
Electron volt (eV)	1eV=1.60×10^{-19} J

Select five out of following six problems.



1 (20pts). A nonconducting sphere of radius R carries a uniform charge per unit volume ρ. Let $\boldsymbol{r}$ be the vector from the center of the sphere to a point P.

a. Show that the electric field at P is given by $\boldsymbol{E} = \rho\ \boldsymbol{r}/3\varepsilon_0$ when the point P is within the sphere.

b. Show that the electric field at P is given by $\boldsymbol{E} = \rho R^3\boldsymbol{r}/3\varepsilon_0 r^3$ when the point P is outside the sphere.

c. A spherical cavity is created in the sphere, as shown in the accompanying figure. In this case, show that the electric field at all points within the cavity is $\boldsymbol{E} = \rho\ \boldsymbol{a}/3\varepsilon_0$ (uniform field), where $\boldsymbol{a}$ is the vector connecting the center of the charged sphere and the center of the cavity.

2.(20pts). A uniformly charged rod of length L with linear charge density λ is placed on the z-axis from $z=-L/2$ to $z=L/2$.

(1) Obtain the electrical potential $V(x,y)$ for any point on the x-y plane.

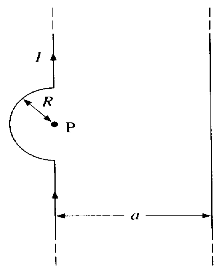
(2) What would you expect the potential to be for $\sqrt{x^2+y^2} >> L$?

(3) What is the electric field in the x-y plane in the limit that $\sqrt{x^2+y^2} << L$?

3. (20pts). Two long wires, one of which has a semicircular bend of radius R, are positioned as shown in the accompanying figure. Both wires carry a current I.

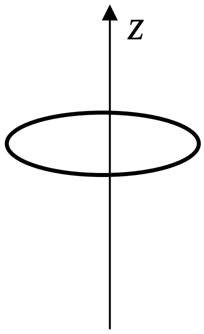
(a) How far apart must they be so that the resultant magnetic field at the point P is zero?

(b) Does the current in the straight wire flow up or down?



4.(20pts). A circular wire loop of radius r falls under the influence of gravity and in the presence of a magnetic field given by

$$\boldsymbol{B} = B_0(1+az)\boldsymbol{e}_z - \frac{aB_0\rho}{2}\boldsymbol{e}_\rho$$



in the cylindrical coordinates, here a is a constant. The loop remains horizontal while falling. The wire has the mass m and the resistance of the loop is R.

(a) Find the induced emf in the loop when the loop falls down with the velocity v.

(b) Assuming that $a > 0$, determine the direction of the induced current. Explain.

(c) Show that the terminal velocity of the loop is $v_t = \dfrac{mgR}{a^2 B_0^2 \pi^2 r^4}$.

5. (20pts). A CD (thin insulating disk with a hole in the middle, inner radius a, out radius b) spins counter-clockwise in the x-y plane with angular velocity ω. The mass of the CD disk is m. A total charge Q is distributed uniformly on the surface of the disk.

(a) Find the magnetic moment of the spinning disk.

(b) Find the angular momentum and the the gyromagnetic ratio of the spinning disk.

(c) Suppose a uniform magnetic field B is applied in the y-direction. What is the angular velocity with which the disk will precess?

6. (20pts) A circular parallel-plate capacitor of radius R and plate separation d is connected to a source of potential difference $V = V_0 \cos \omega t$. The edge effects can be ignored since $d << R$.

(a) Find the electric field inside the capacitor and the area densities of the electric charge on the plates as a function of time.

(b) Find the magnetic field inside the capacitor as a function of time and the distance to the axis of the plate.

(c) Find the electromagnetic field energy flow density (the Poynting vector) through the lateral surface of the capacitor.

1	2	3	4	5	6	Sum
/20	/20	/20	/20	/20	/20	/100

Name ______________________ Nov. 11, 2005 Sunday

Department _________________

SID _______________________

University Physics

Midterm Examination

School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

Electron charge	$e = 1.60\times10^{-19}$ C
Dielectric constant of free space	$\varepsilon_0 = 8.85\times10^{-12}$ F/m
Permeability of free space	$\mu_0 = 4\pi\times10^{-7}$ N/A^2
Vacuum speed of light	$c = 3.00\times10^{8}$ m/s
Electron mass	$m_e = 9.11\times10^{-31}$ kg
Proton mass	$m_p = 1.67\times10^{-27}$ kg
Electron volt (eV)	1eV=1.60×10^{-19} J

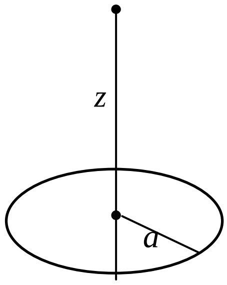
Select five out of following six problems.

1. The electric charge density of a spherically symmetric charge distribution is

$$\rho(r) = \begin{cases} \rho_0(1-\dfrac{r^2}{a^2}), & r < a, \\ 0\,, & r > a, \end{cases} \qquad (a > 0).$$

(a) Calculate the total charge Q.
(b) Find the electric field $\boldsymbol{E}$ and the electric potential $V(r)$ in the region $r > a$.
(c) Find the electric field $\boldsymbol{E}$ and the electric potential $V(r)$ in the region $r < a$.

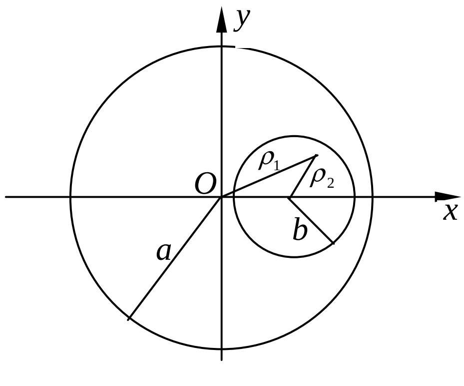
2. A circular disk of radius a has the surface charge density σ.



(a) Compute the electric potential at a point on the axis of symmetry and at distance z from the center of the disk and then the electric field there.

(b) Find the discontinuity of the electric field at the center of the disk. Reobtain the result by using the Gauss's law of the electric field.

3. A steady electric current I in the z-direction flows uniformly in the region between two cylindrical surfaces $x^2+y^2=a^2$ and $(x-a/2)^2+y^2=b^2$, where $a>0$, $b>0$ and $b<a/2$.



(a) Find the magnitude and direction of the magnetic field $\boldsymbol{B}$ at every point P on the x-axis.

(b) Show that the magnetic field $\boldsymbol{B}$ is uniform in the region $(x-a/2)^2+y^2<b^2$ and the field is

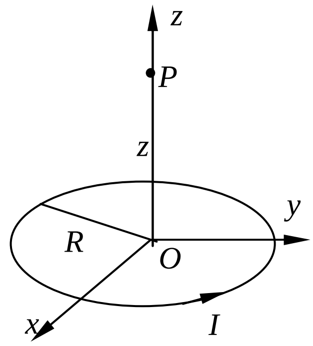
$$\boldsymbol{B}=\frac{\mu_0 I}{4\pi}\frac{a}{a^2-b^2}\boldsymbol{e}_y .$$

(Hint: A vector in the x-y plane that is perpendicular to the displacement $\boldsymbol{\rho}$ in the x-y plane and of the magnitude $\rho=|\boldsymbol{\rho}|$ can be expressed as $\pm\boldsymbol{e}_z\times\boldsymbol{\rho}$.)

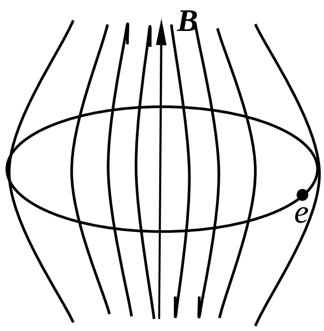
4. As shown in the accompanying figure, a loop of radius R carries an electric current I.

(a) Find the magnetic field at point P that is on the symmetric axis of the loop and at distance z from the center of the loop.

(b) If a magnetic dipole with magnetic moment $\boldsymbol{m} = m_0 \boldsymbol{e}_z$ is placed at point P, what is the magnitude and direction of the force acted on the dipole?



5. The betatron is a cyclic induction electron accelerator. In a betatron, the electron is accelerated by the circular electric field induced by the changing magnetic field that is axially symmetric. Simultaneously, the magnetic field also provides the centripetal force for the electron to move in a circular orbit. The poles of the magnet are tapered to cause the axially symmetric magnetic field to weaken with increasing radius. Show that if the average magnetic field inside the orbit



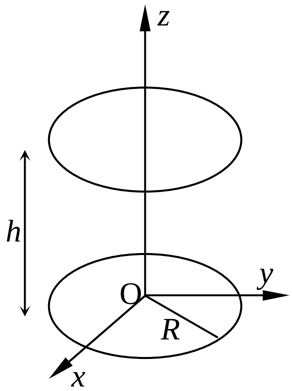
$$\overline{B} = \frac{1}{\pi R^2} \int_0^R B(\rho) 2\pi\rho d\rho$$

is always twice as strong as the magnetic field on the orbit $B(R)$, i.e., $\overline{B} = 2B(R)$, the electron can be accelerated stably on the orbit with fixed radius R.

6. A non-uniform time-varying electric field $\boldsymbol{E}$ given by

$$\boldsymbol{E} = E_0\left(1-\frac{\rho}{R}\right)\sin(\omega t)\,\boldsymbol{e}_z$$

is distributed in the cylindrical region of radius R and height h as shown in the figure.



(a) Find the displacement current density j_d in the cylindrical region ($\rho < R$).

(b) Find the magnetic field $\boldsymbol{B}$ induced in the same region.

(c) Indicate on the figure the direction of $\boldsymbol{B}$ at radial distance ρ from the axis at $t = 0$.

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________________ November 6, 2016 Sunday

Department/School:____________________ SID:__________________________________

University Physics II

Midterm Examination

Kuang Yaming Honors School, Nanjing University

<u>Select five out of following six problems.</u>

1.(20pts.) Three point charges $2q$, q and $2q$ are restricted to the closed interval $-a \le x \le a$ on the x-axis.

(a) Determine the positions where the charges should be placed so that the potential energy of the system is a minimum.

(b) Supposing that the three point charges are located at the positions as determined in (a), find the electrical potential $V(x)$ at the point x on the x-axis where $|x| >> a$, and approximate the result by expanding it up to the second order of $a/|x|$.

(c) For the same arrangement of the point charges in (b), find the approximate electrical field $\boldsymbol{E}(x)$ at the point x on the x-axis (where $|x| >> a$) up to the second order of $a/|x|$.

2.(20pts.) A pair of magnetic monopoles are placed at $\boldsymbol{r} = \boldsymbol{a}$ and $\boldsymbol{r} = -\boldsymbol{a}$, with the corresponding magnetic charges being g and $-g$, respectively.

(a) According to Dirac's hypothesis, write down the magnetic induction $\boldsymbol{B}$ at the point $\boldsymbol{r}$ ($\boldsymbol{r} \neq \pm\boldsymbol{a}$).

(b)The configuration can also be taken as a magnetic dipole when viewed from the far zone ($r >> a, a = |\boldsymbol{a}|$), and the dipole moment can be defined as $\boldsymbol{p} = 2g\boldsymbol{a}$. Find the magnetic induction $\boldsymbol{B}$ produced by this dipole by expanding the expression of (a) up to the first order of a/r.

(c) Suppose there is another magnetic dipole moment $\boldsymbol{\mu}$ that is constituted of a small electric current loop in the magnetic field $\boldsymbol{B}$ of (b). The dipole moment $\boldsymbol{\mu}$ is located at the position $\boldsymbol{r}$ within the far zone. Find the torque $\boldsymbol{M}$ acted on the magnetic dipole $\boldsymbol{\mu}$ if $\boldsymbol{r} = x\,\boldsymbol{i}$, $\boldsymbol{\mu} = \mu\,\boldsymbol{j}$ and $\boldsymbol{a} = a\,\boldsymbol{k}$ where $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ are the three base vectors of Cartesian system.

(Hint: $f(\boldsymbol{r}+\boldsymbol{a}) = f(\boldsymbol{r}) + \boldsymbol{a}\cdot\nabla f(\boldsymbol{r}) + \cdots$, and in particular $\dfrac{1}{|\boldsymbol{r}+\boldsymbol{a}|^3} = \dfrac{1}{r^3} + \boldsymbol{a}\cdot\nabla\dfrac{1}{r^3} + \cdots$ and $\nabla\dfrac{1}{r^3} = -\dfrac{3\boldsymbol{r}}{r^5}$.)

3. (20pts.) Three thin cylindrical shells are coaxially fixed together; their radii are R, $2R$, and $4R$, respectively. The inner and outer shells are both metallic and are both connected to the ground; the middle one is plastic, with areal charge density σ.

(a) Find the areal charge density σ_1 of the inner shell.

(b) Find the electric field $\boldsymbol{E}_1$ between the inner and the middle shells, and $\boldsymbol{E}_2$ between the middle and the outer ones.

4. (20pts.) Two parallel plates enclose a dielectric material that has a varying relative dielectric function: $\varepsilon_r = \varepsilon_r(x) = e^{\alpha x}$ where $\alpha > 0$ is a constant and x is the distance apart from the lower plate. Suppose that the areal charge densities of the upper and lower plates are σ and $-\sigma$ respectively.

(a) What is the electric displacement $\boldsymbol{D}$ within the medium?

(b) What is the electric field $\boldsymbol{E}$ within the medium?

(c) If the two plates are of area S, and a distance d apart, find the energy U stored in this configuration.

(d) Find the volume charge density ρ distributing within the medium.

5. (20 pts.) A long solenoid of radius a, carrying n turns per unit length, is looped perpendicularly from outside by a closed conducting wire of electrical resistance R.

(a) If the current I in the solenoid is increasing at a constant rate ($dI/dt = k$), what electrical current J flows in the loop?

(b) If the current I in the solenoid is keeping constant but the solenoid is pulled out of the wire loop, turned around, and reinserted into it perpendicularly, what total charge Q that passes through the conducting loop during the process.

6. (20pts.) In free space, there exists a traveling electromagnetic wave $\boldsymbol{B}(x, y, z) = B_0 \cos(kz - \omega t)\, \boldsymbol{e}_y$ where $\boldsymbol{B}$ is the magnetic induction. Here, B_0 is a constant, x, y and z are Cartesian coordinates, and $\boldsymbol{e}_x$, $\boldsymbol{e}_y$ and $\boldsymbol{e}_z$ the corresponding unit vectors, t denotes the time, k the wave vector, and ω the frequency: $\omega = ck$ where c is the speed of light.

(a) Show that $\nabla \cdot \boldsymbol{B} = 0$.

(b) Supposing that the electric field $\boldsymbol{E}$ takes on the form: $\boldsymbol{E} = \boldsymbol{E}_0 \cos(kz - \omega t)$ where $\boldsymbol{E}_0$ is a constant vector, find $\boldsymbol{E}_0$ by the Maxwell equation $\nabla \times \boldsymbol{B} = \dfrac{1}{c^2}\dfrac{\partial}{\partial t}\boldsymbol{E}$.

(c) Find the Poynting vector $\boldsymbol{S}$ and its average $\langle \boldsymbol{S} \rangle$ over one period of time.

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________________ November 19, 2017 Sunday

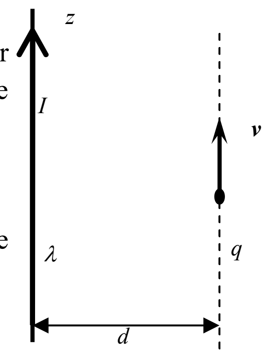
Department/School:____________________ SID:__________________________________

University Physics II

Midterm Examination

Kuang Yaming Honors School, Nanjing University

<u>Select five out of following six problems.</u>



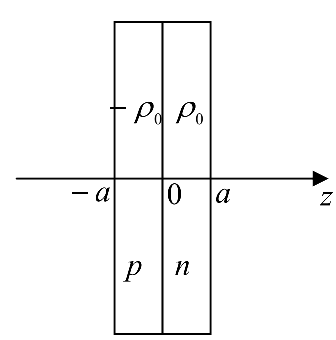
1. (20pts.) As shown in the accompanying figure, an infinitely long straight wire with a uniform linear charge density λ lies along the z-axis. The wire also carries an electric current I that flows in the positive direction of z-axis.

(a) What is the electric field $\boldsymbol{E}$ around the wire?

(b) What is the magnetic field $\boldsymbol{B}$ around the wire?

(c) A particle of charge q is a distance d away from the wire and is travelling in a straight line parallel to the wire. What is the velocity of the particle?

2.(20pts.) When two slabs of p-type and n-type semiconductors are put in contact, the charge carriers in the p-region (holes with positive charge) tend to diffuse to n-region and the charge carriers in the n-region(electron with negative charge) tend to diffuse to the p-region. Therefore, at the interface of two types of semiconductors, an inversion layer is formed in which a volume in the n-type material is positively charged while a volume in the p-type material is negatively charged. Assume that electric charge density in the inversion layer is



$$\rho(x,y,z)=\begin{cases}\rho_0 & 0<z<a\\ -\rho_0 & -a<z\le 0\\ 0 & |z|\ge a\end{cases}.$$

(a) Find the electric field everywhere.

(b) By setting the zero potential point at $z=0$, find the electric potential everywhere.

(Hint: The inversion layer is thin enough so that the areas of slabs can be approximately taken as infinities.)

3. (20pts.) Two square parallel-plate electrodes of sides l are a distance d apart. A dielectric slab of permittivity ε, length l, width b $(b < l)$ and thickness d is inserted between the plates, with three of four side surfaces being exactly coincident with the three of four edges of the electrodes.

(a) Find the capacitance C of the system.

(b) Find the electrostatic energy U stored in the system after the electrodes are connected to a battery of voltage V for a long enough time.

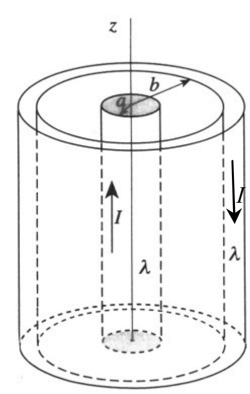
(c) The capacitor is charged as in (b) and then the battery is disconnected. What is the force acted on one of the plate now? Is the force attractive or repulsive one?

4. (20pts.) Suppose, according to Dirac's hypothesis, there exists a sphere of radius R that is fulfilled uniformly with amount of magnetic charge (magnetic monopole) G.

(a) Find the magnetic induction $\boldsymbol{B}$, both outside and inside the sphere.

(b) Find the energy of the magnetic field.

5. (20pts.) The infinitely long coaxial cable carries a steady electric current I upward in the inner conductor of radius a and a return current I downward in the out conducting shell of inner radius b, as shown in the figure. The resistances per unit length of two conductors are both λ. The space between two conductors $(a < \rho < b)$ is filled with vacuum. The electric potential in the vacuum space $(a < \rho < b)$ is calculated as



$$V(\rho, z) = -I\lambda z \frac{\ln\left(\frac{\rho^2}{ab}\right)}{\ln\left(\frac{a}{b}\right)}$$

(a) Find the electric field $\boldsymbol{E}$ in the region $a < \rho < b$.

(b) Find the magnetic field $\boldsymbol{H}$ in the region $a < \rho < b$.

(c) Find the Poynting vector $\boldsymbol{S}$ in the region $a < \rho < b$.

(d) Calculate the energies per unit time flow into per unit length of two conductors respectively. Explain your results.

(Hint: In the cylindrical coordinates $\operatorname{grad} V = \frac{\partial V}{\partial \rho}\boldsymbol{e}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\boldsymbol{e}_\phi + \frac{\partial V}{\partial z}\boldsymbol{e}_z$.)

6. (20pts.) Suppose that in a neutral $(\rho = 0)$ and conducting medium, the physical quantities of electromagnetic field obey the linear constitutive relations $\boldsymbol{D} = \varepsilon \boldsymbol{E}$, $\boldsymbol{B} = \mu \boldsymbol{H}$ and $\boldsymbol{j} = \sigma \boldsymbol{E}$ (Ohm's law in differential form), where ε, μ and σ are all positive constants.

(a) Write the four Maxwell equations of $\boldsymbol{E}$ and $\boldsymbol{B}$ only in differential forms, by using the above constitutive relations.

(b) Show that the electric field $\boldsymbol{E}$ satisfies the equation $\nabla^2 \boldsymbol{E} - \mu\sigma\, \partial \boldsymbol{E}/\partial t - \mu\varepsilon\, \partial^2 \boldsymbol{E}/\partial t^2 = 0$.

(b) Suppose the electric field $\boldsymbol{E}$ takes the form of wave solution $\boldsymbol{E} = E_0 \boldsymbol{e}_x \exp[i(kz - \omega t)]$, where ω is the real angular frequency and k is the complex wave number. Find the equation satisfied by ω and k.

(Hint: $\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}$)

University Physics II - 3
Midterm Examination
Kuang Yaming Honors School, Nanjing University
<u>Select five out of following six problems.</u>

1.There are three arges, q, q and $-q$, which are all placed on x-axis. The coordinates for them are $x = a$, $x = 0$, and $x = -a$, respectively. For thefar zone, $|x| >> a$, and to the order of magnitude of dipole or a / x, what is the electric potential ϕ and field $\boldsymbol{E}$ on the point x?

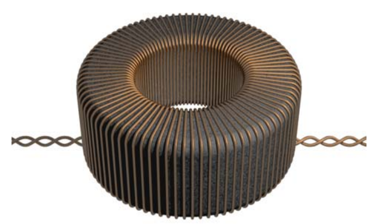
2.A long cylinder of radiu R carries a charge density, $\rho = k\ r$, where k is a constant and r is the distance from the axis.

(1) Find the electric field E inside and outside the cylinder.

(2) Now, let the cylinder be shielded coaxially with a metal shell of radius R' find the energy U of the electric field per unit length.

3. A parallel late capacitor has area A and separation d. The upper half of the capacitor is filled with an insulator of relative dielectric constant ε_1, and the lower half with another insulator of relative dielectric constant ε_2. (1) Find the capacitance C of the capacitor. (2) Find the areal charge density σ of the interface between the two insulators if the capacitor is charged to a voltage V.

4. Find the self-inductanc L of a toroidal coil with rectangular cross section (inner radius a, outer radius b, height h), which carries a total of N turns.



5. A soap bubble of r ius r is in equilibrium with the outer air of pressure p. Now, it is slowly given a charge q, the radius of the bubble will increase gradually to R, due to the mutual repulsion of the surface charge. Regard the air as an ideal gas, and neglect the effect of the tension of the surface of the bubble.

(1) Find the pressure p_{field} of the electric field upon the bubble, the dielectric constant of the outer air being ε.

(2) Show that $q^2 = 32\pi^2 \varepsilon p R (R^3 - r^3)$.

6.Ther are two electromagnetic travelling waves overlapped in free space, their electric fields are $\boldsymbol{E}_1 = C\cos(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)\,\boldsymbol{e}_x$ and $\boldsymbol{E}_2 = D\cos(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)\,\boldsymbol{e}_x + D\cos(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)\,\boldsymbol{e}_y$, where $C > 0$ and $D > 0$ are constant, and ω and $\boldsymbol{k} = k\,\boldsymbol{e}_z$ are the frequency and wave vector ($\omega = ck$ where c is the speed of light), respectively. Here, x, y and z are Cartesian coordinates, and $\boldsymbol{e}_x$, $\boldsymbol{e}_y$ and $\boldsymbol{e}_z$ the corresponding unit vectors, t denotes the time, and $\boldsymbol{r} = x\,\boldsymbol{e}_x + y\,\boldsymbol{e}_y + z\,\boldsymbol{e}_z$.

(1) Find the total magnetic induction $\boldsymbol{B}$.

(2) Find the energy density u of the electromagnetic field and its time average $\langle u \rangle$ over one period.

(3) Find the Poynting vector $\boldsymbol{S}$ and its time average $\langle \boldsymbol{S} \rangle$ over one period.

University Physics II - 5
Midterm Examination
Kuang Yaming Honors School, Nanjing University
Select five out of following six problems.

1. (20 pts.) Find the electric field (**E** magnitude and direction) a distance $z=$ above the midpoint of a straight line segment of length L, which carries a uniform line charge λ . Check that your result is consistent with what you would expect when z>>L.

2. (20 pts.) Suppose that there exists a magnetic monopole g that is located at the origin and a magnetic dipole with the magnetic dipole moment of $\boldsymbol{m}$ is placed on the x-axis at $x=a$ $(a > 0)$.

(a) What is the magnetic field $\boldsymbol{B}$ produced by the monopole at the position of the dipole?
(b) What is the potential energy of the dipole in the magnetic field of the monopole?
(c) If the dipole moment $\boldsymbol{m}$ is anti-parallel to the x-axis, find the force $\boldsymbol{f}$ acted on it.

3.(20 pts.) A long cylindrical shell of length L has an inner radius a and an outer radius b is made of a material of electrical conductivity σ. The inner and outer surfaces of the shell are maintained at a potential difference V. (The two cylindrical surfaces are equipotential surfaces.)

(a) What is the electric field inside the shell ($b < \rho < a$)?

(b) What is the electric current that flows from the inner surface to the outer surface?

Solution1 (b). The electrical resistance is

4.(20 pts.) A metal sphere of radius R that carries electric charges Q is insulated from the air with a spherical shell of

medium of relative dielectric constant ε_r. The outer radius of

the dielectric medium is R'. The relative dielectric constant of the air can be taken as 1.0 .

(a) Find the electric field $\boldsymbol{E}$ outside the metal sphere.
(b) Find the areal charge density σ on the outer surface of the dielectric shell.
(c) What is the capacitance C of this configuration?

5. (20 pts.) Consider a long coaxial cable in which the inner conductor

is a cylinder of radius a made of material having electrical resistivity ρ and relative magnetic permeability μ and the out shield is a perfect conductor in the shape of a thin cylindrical shell of radius b. The inner conductor is connected (shorted) to the out shield at the right end and a voltage V_0 is applied at the left end of the cable (as shown in the figure) so that a steady current is established after some time. Assume that the electric current densities are uniform in the inner conductor and in the out shield of the cable and the length of the cable is large enough ($l >> b > a$) so that the edge effect can be neglected.

(a) Calculate the magnetic field everywhere within the coaxial cable.
(b) Find the energy of the magnetic field.
(c) Find the self-inductance of the coaxial cable.

6. (20 pts.) An electromagnetic wave travels in free space. The electric field is $\boldsymbol{E} = E_0\,[\cos(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)\,\boldsymbol{e}_x + \sin(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)\,\boldsymbol{e}_y]$, where $E_0 > 0$ is a constant, and ω and $\boldsymbol{k} = k\,\boldsymbol{e}_z$ are the frequency and wave vector ($\omega = c\,k$ where c is the speed of light), respectively. Here, x, y and z are Cartesian coordinates, $\boldsymbol{e}_x$, $\boldsymbol{e}_y$ and $\boldsymbol{e}_z$ the corresponding unit vectors, t denotes the time, and $\boldsymbol{r} = x\,\boldsymbol{e}_x + y\,\boldsymbol{e}_y + z\,\boldsymbol{e}_z$.

(a) What is the spinning direction of $\boldsymbol{E}$ at a fixed point $\boldsymbol{r}$, clockwise or anticlockwise?

(b) Verify that the electric field $\boldsymbol{E}$ satisfies the equation $\nabla^2\boldsymbol{E} - \frac{1}{c^2}\frac{\partial^2\boldsymbol{E}}{\partial t^2} = 0.$

(c) Find the magnetic induction $\boldsymbol{B}$.

(d) Find the Poynting vector $\boldsymbol{S}$.